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Time : 3.00 hrs.

Quarterly Examination - 2025
MATHEMATICS

Reg. No. Masiya

Max. Marks : 90

PART - I

Choose the most appropriate answer from the given four alternatives and write the option code and answer.

1. If $P = \begin{bmatrix} 1 & x & 0 \\ 1 & 3 & 0 \\ 2 & 4 & -2 \end{bmatrix}$ is the adjoint of 3 x 3 matrix A and $|A| = 4$, then x is a) 15 b) 12 c) 14 d) 11

2. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then λ is a) 17 b) 14 c) 19 d) 21

3. If $\rho(A) = \rho(A|B)$, then the system Ax = B of linear equations is
a) consistent and has a unique solution b) consistent c) consistent and has infinitely many solution d) inconsistent

4. If $|z - 2 + i| \le 2$, then the greatest value of $|z|$ is a) $\sqrt{3} - 2$ b) $\sqrt{3} + 2$ c) $\sqrt{5} - 2$ d) $\sqrt{5} + 2$

5. If z = x + iy is a complex numebr such that $|z + 2| = |z - 2|$, then the locus of z is
a) real axis b) imaginary axis c) ellipse d) circle

6. The product of all four values of $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{\frac{3}{4}}$ is a) –2 b) –1 c) 1 d) 2

7. The value of the complex number $(i^{25})^3$ is equal to a) 1 b) i c) –i d) –1

8. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k satisfies a) $|k| \le 6$ b) k = 0 c) $|k| > 6$ d) $|k| \ge 6$

9. The polynomial $x^3 + 2x + 3$ has a) one negative and two imaginary zeros b) one positive and two imaginary zeros
c) three real zeros d) no zeros

10. A polynomial equation of degree n always has
a) exactly 'n' roots b) n distinct roots c) 'n' real roots d) n imaginary roots

11. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ d) $\tan^{-1}\left(\frac{1}{2}\right)$

12. If $\text{Cot}^{-1}x = \frac{2\pi}{5}$ for some $x \in R$, then the value of $\tan^{-1}x$ is a) $\frac{-\pi}{10}$ b) $\frac{\pi}{5}$ c) $\frac{\pi}{10}$ d) $\frac{-\pi}{5}$

13. $\text{Sin}^{-1}(2\cos^2 x - 1) + \cos^{-1}(1 - 2\sin^2 x) =$ a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{6}$

14. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

15. If the normals of the parabola $y^2 = 4x$ drawn at the end points of the latus rectum are tangent to the circle $(x-3)^2 + (y+2)^2 = r^2$ then the value of r^2 is a) 2 b) 3 c) 1 d) 4

16. The circle passing through (1, –2) and touching the axis of x at (3, 0) passing through the point
a) (–5, 2) b) (2, –5) c) (5, –2) d) (–2, 5)

17. If $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$, then the value $[\vec{a}, \vec{b}, \vec{c}]$ of is a) $|\vec{a}||\vec{b}||\vec{c}|$ b) $\frac{1}{3}|\vec{a}||\vec{b}||\vec{c}|$ c) 1 d) –1

18. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, non-zero vectors such that $[\vec{a}, \vec{b}, \vec{c}] = 3$, then $[[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}]]^2$ is equal to
a) 81 b) 9 c) 27 d) 18

19. If the planes $\vec{r} \cdot (2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ are parallel, then the value of λ and μ are
a) $\frac{1}{2}, -2$ b) $\frac{-1}{2}, 2$ c) $\frac{-1}{2}, -2$ d) $\frac{1}{2}, 2$

20. If $\lambda\vec{i} + 2\lambda\vec{j} + 2\lambda\vec{k}$ is a unit vector, then the value of λ is a) $\pm\frac{1}{3}$ b) $\pm\frac{1}{4}$ c) $\pm\frac{1}{9}$ d) $\frac{1}{2}$

PART - II

Answer any seven of the following question. Question no.30 is compulsory. 7 x 2 = 14

21. Find the rank of the matrix $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$

22. Write in polar form of the complex number –2 – i2.

23. Find the modulus of the complex number 2i (3 – 4i) (4 – 3i) Symbol

24. Find a polynomial equation of minimum degree with rational co-efficients, having 2i + 3 as a root. (S R) (P E)

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25. Construct a cubic equation with roots 1, 1 and –2.

26. Find the principal value of $\sin^{-1}\left(\frac{-1}{2}\right)$

27. Identify the type of conic section for the equation $3x^2 + 2y^2 = 14$.

28. Find the general equation of a circle with centre (–3, –4) and radius 3 units.

29. Show that the vectors $\hat{i}+2\hat{j}-3\hat{k}$, $2\hat{i}-\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ are coplanar.

30. Find the parametric form of vector equation of a line passing through a point (2, –1, 3) and parallel to line $\vec{r} = (\hat{i}+\hat{j}) + t(2\hat{i}+\hat{j}-2\hat{k})$

PART - III

Answer any seven of the following questions. Q.No.40 is compulsory. **7 x 3 = 21**

31. Solve the following system of linear equations, using matrix inversion method. $5x + 2y = 3$; $3x + 2y = 5$

32. Solve the following system of linear equations by Cramer's rule : $\frac{3}{x}+2y=12$; $\frac{2}{x}+3y=13$

33. Simplify $\left(\sin\frac{\pi}{6}+i\cos\frac{\pi}{6}\right)^{18}$

34. Find the square roots of 4 + 3i.

35. Solve the cubic equation $2x^3 - x^2 - 18x + 9 = 0$ if sum of two of its roots vanishes.

36. Find the value of $\sin^{-1}\left(\sin\frac{5\pi}{9}\cos\frac{\pi}{9}+\cos\frac{5\pi}{9}\sin\frac{\pi}{9}\right)$

37. A particle acted on by constant forces $8\hat{i}+2\hat{j}-6\hat{k}$ and $6\hat{i}+2\hat{j}-2\hat{k}$ is displaced from the point (1, 2, 3) to the point (5, 4, 1). Find the total work done by the forces.

38. Find the equation of the parabola whose vertex is (5, –2) and focus (2, –2)

39. Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$

40. Show that the distance from the origin to the plane $3x + 6y + 2z + 7 = 0$ is 1.

PART - IV

Answer all the questions. **7 x 5 = 35**

41. a) Solve the following system of linear equation by Cramer's rule. $\frac{3}{x}-\frac{4}{y}-\frac{2}{z}-1=0$, $\frac{1}{x}+\frac{2}{y}+\frac{1}{z}-2=0$ $\frac{2}{x}-\frac{5}{y}-\frac{4}{z}+1=0$

(OR)

b) Investigate the values of λ and μ the system of linear equations $2x + 3y + 5z = 9$, $7x + 3y - 5z = 8$, $2x + 3y + \lambda z = \mu$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions

42. a) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right)=\frac{\pi}{4}$, show that $x^2 + y^2 + 3x - 3y + 2 = 0$ **(OR)**

b) Solve the equation $z^3 + 8i = 0$, where $z \in C$.

43. a) Solve $(x - 5)(x - 7)(x + 6)(x + 4) = 504$ **(OR)**

b) Solve the equations $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$

44. a) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, show that $x + y + z = xyz$. **(OR)**

b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$.

45. a) Find the centre, foci and eccentricity of the hyperbola $11x^2 - 25y^2 - 44x + 50y - 256 = 0$ **(OR)**

b) At a water fountain, water attains a maximum height of 4m at horizontal distance of 0.5 m from its origin. If the path of water is a parabola, find the height of water at a horizontal distance of 0.75 m from the point of origin.

46. a) Prove by Vector method that $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$ **(OR)**

b) Find the points where the straight line passes through (6, 7, 4) and (8, 4, 9) cuts the xz and yz planes.

47. a) Find the non-parametric form of vector equation and Cartesian equation of the plane passing through the point (1, –2, 4) and perpendicular to the plane $x + y - 3z = 11$ and parallel to the line $\frac{x+7}{3}=\frac{y+3}{-1}=\frac{z}{1}$ **(OR)**

b) If z_1, z_2 and z_3 are three complex number such that $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$

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